VitaMotus.earth

Project Vision Guide - Developer Handoff

Mission Statement

VitaMotus.earth is a calm, educational spider archive designed to help visitors understand how spiders are classified, how their bodies are built, where they live, and why they matter. The first build should start with spiders only and should be structured around a clear taxonomy backbone: Family -> Genus -> Species. From my end, the main data input will be a database where each species record carries its family, genus, scientific name, common name where reliable, authority, year, distribution, habitat, size, behaviour, venom note, conservation status, interesting fact, photo credits, and references.

The developer should understand VitaMotus as a living archive, not a fixed brochure. The website must allow the database to grow over time as new species profiles are added, corrected, sourced, and enriched.

Taxonomy input from my end The core database begins with all spider families, the genera under each family, and the species under each genus. This must drive browse pages, search, filters, and individual species profiles.	**Species of the Day dynamic card** The homepage should show a dynamic Species of the Day card that changes every 24 hours from the databank. It should display the scientific name, common name, image, family, distribution, habitat, and one short interesting note.
Individual species viewing card Every species should have its own clean profile page using a fixed standard: scientific name, common name, family, genus, authority, year described, distribution, habitat, size, behaviour, venom note, conservation status, interesting fact, photo credits, and references.	**Spider anatomy and terminology** The site should include learning pages for spider anatomy and terminology. These pages should explain terms such as cephalothorax/prosoma, abdomen/opisthosoma, pedipalps, chelicerae, fangs, spinnerets, book lungs, eyes, and legs in beginner-friendly language.

0. Document Flow

This guide is organised so a developer can move from purpose, to design, to pages, to data, to implementation. Every heading starts on a fresh page for clean review and printing.

1. Core Vision and Scope

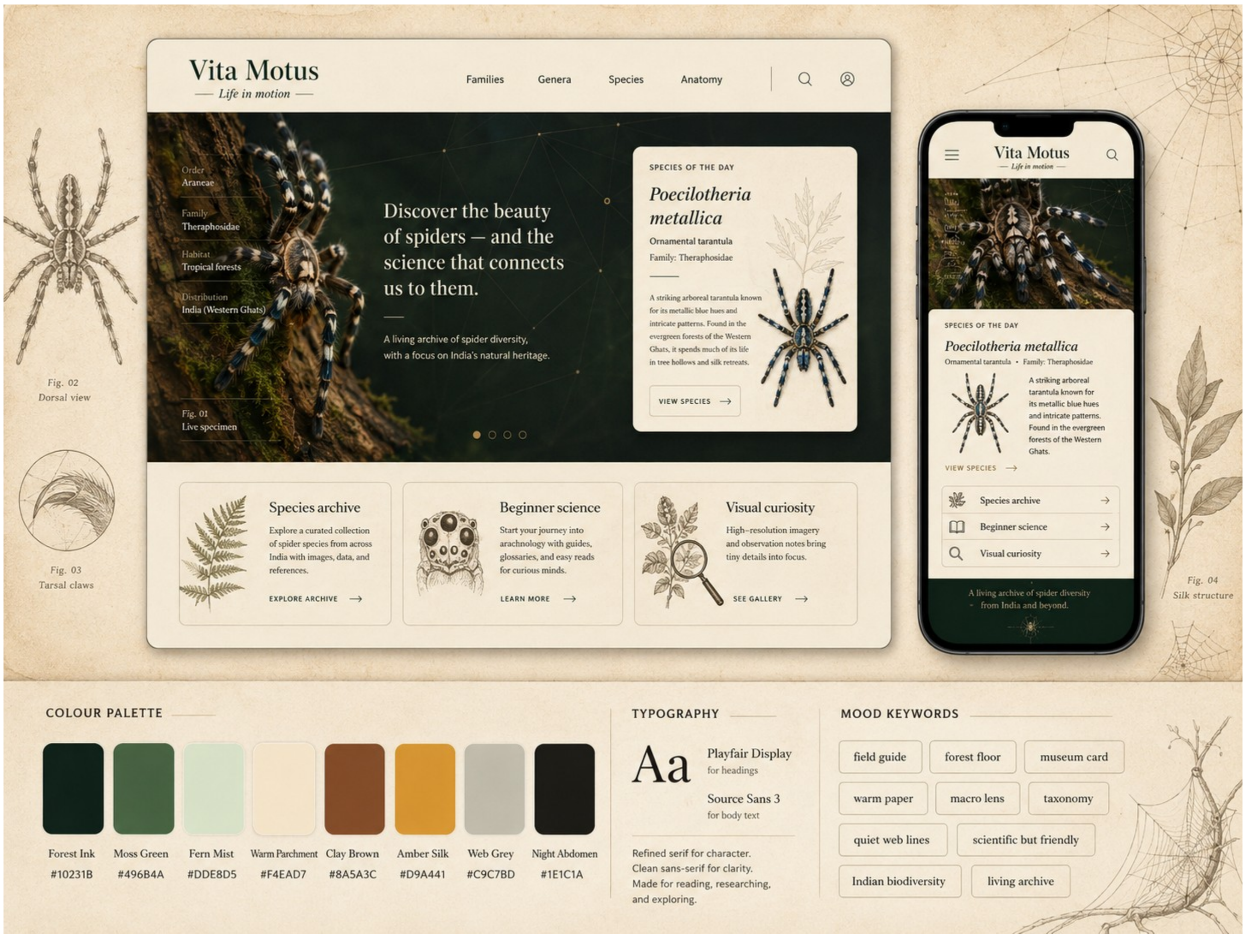
VitaMotus should feel like a living natural-history archive: beginner-friendly, scientifically respectful, visually beautiful, and useful as a growing spider database.

- Start with spiders only. The first public identity should be focused and not diluted across all animals.
- Build the site around a taxonomy skeleton: Family -> Genus -> Species.
- Make every species page expandable, so a basic taxonomy placeholder can later grow into a full educational profile.
- Write for serious hobbyists, students, nature lovers, and visitors who are curious but not academic specialists.
- Use simple language, but keep scientific names, sources, and taxonomy standards visible.

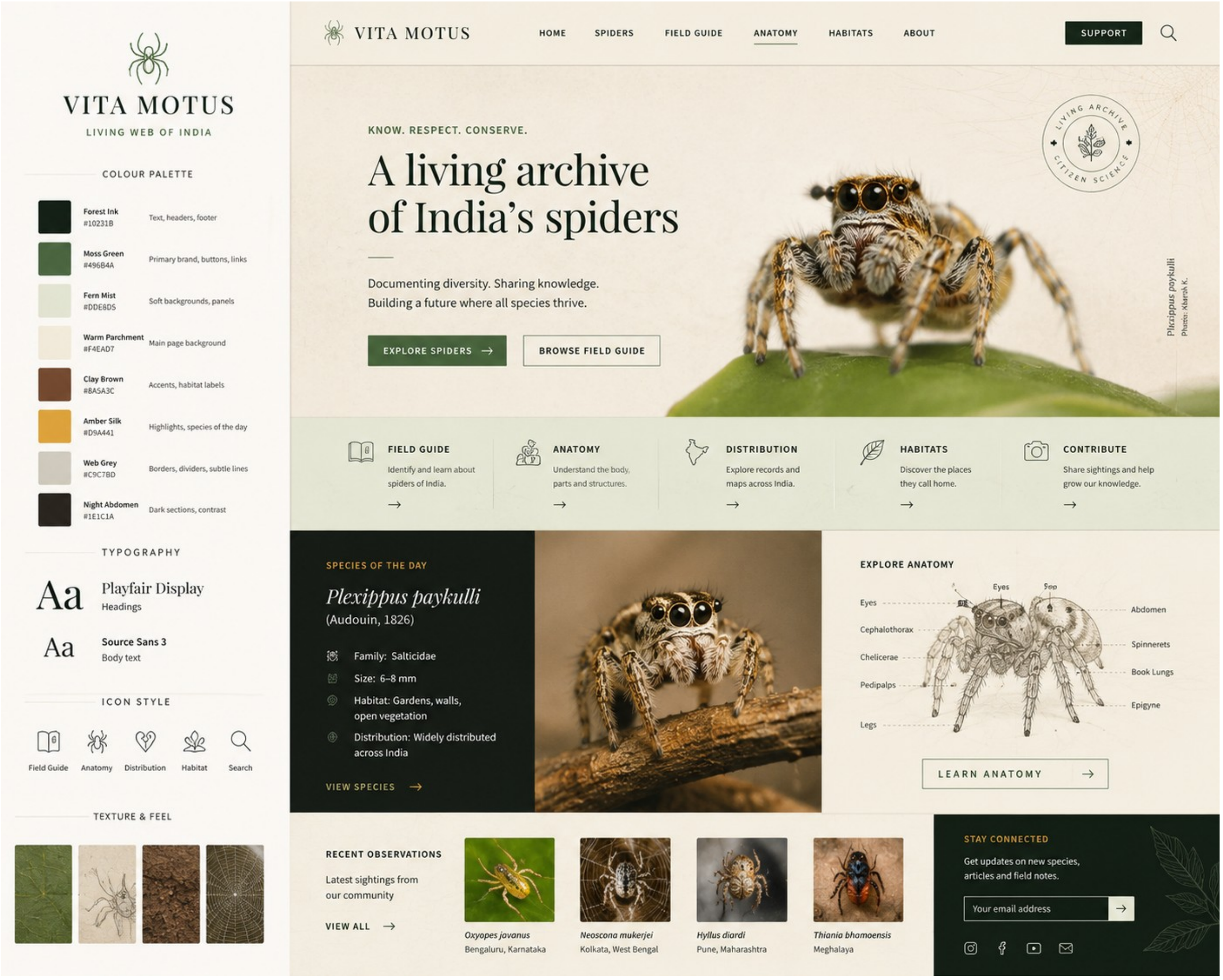
Area	Direction
Primary goal	Create a clean spider archive where visitors can browse families, genera, and species.
Learning goal	Explain anatomy, taxonomy, terminology, behaviour, habitats, venom notes, and conservation in a calm way.
Content rhythm	Allow 10-20 species records to be enriched over time as reading, sourcing, and photo permissions improve.
Tone	Natural-history field guide: educational, warm, precise, and non-horror.

2. Visual Identity and Website Look

The real generated mockups below are the design north star. They show that VitaMotus should look like a polished field-guide website, not a horror spider page or a plain database.



Visual identity board: colour palette, typography, mood keywords, and desktop/mobile direction.



Website look and feel reference: living archive layout, field guide cards, anatomy panel, recent observations, and newsletter area.

Colour role	Use
Warm Parchment	#F4EAD7 - main background
Forest Ink	#10231B - text, headings, footer
Moss Green	#496B4A - primary buttons and links
Clay Brown	#8A5A3C - habitat labels and accents
Amber Silk	#D9A441 - highlights and Species of the Day
Fern Mist	#DDE8D5 - soft panels and taxonomy chips

2A. Homepage Direction

The homepage must immediately explain the archive and give visitors clear entry points into taxonomy, species cards, anatomy, and spider learning.

Homepage direction: hero message, Species of the Day card, search bar, taxonomy chips, and four public learning sections.

- Hero message: explain that VitaMotus explores how spiders are built, behave, and belong in nature.
- Primary actions: Browse species and Learn anatomy.
- Dynamic Species of the Day card: selected from the database every 24 hours.
- Search: allow search across species, genera, families, habitat, and distribution.
- Section cards: Browse Families, Spider Directory, Learn Spider Anatomy, Learn Spider Taxonomy.

3. Website Structure and Public Pages

The site should be navigable before every species profile is complete. A visitor should be able to browse the taxonomy skeleton from day one and later find richer profiles as they are added.

Public page / module	Developer meaning
What is VitaMotus?	Introductory page explaining the archive, the spider-only starting phase, and the educational mission.
Start with spiders	A beginner page explaining what spiders are and how they differ from insects and other arthropods.
Browse Families	List all spider families. Each family page should show accepted genera and species counts and a list of genera.
Browse Genera	List genera alphabetically and by family. Each genus page should show species under that genus.
Browse Species	Searchable species index with filters for family, genus, distribution, habitat, status, and profile completeness.
Species of the Day	Dynamic homepage and archive module that highlights one species every 24 hours.
Learn Spider Anatomy	Interactive or illustrated page explaining external and internal anatomy terms.
Learn Spider Taxonomy	Beginner explanation of order, family, genus, species, authority, year, and changing scientific names.

Recommended public routes: /, /about, /families, /family/{family-name}, /genera, /genus/{genus-name}, /species, /species/{scientific-name}, /anatomy, /taxonomy, /species-of-the-day.

4. Species Page Standard Template

Every species page should use the same standard data structure so that VitaMotus remains consistent, searchable, and easy to maintain.

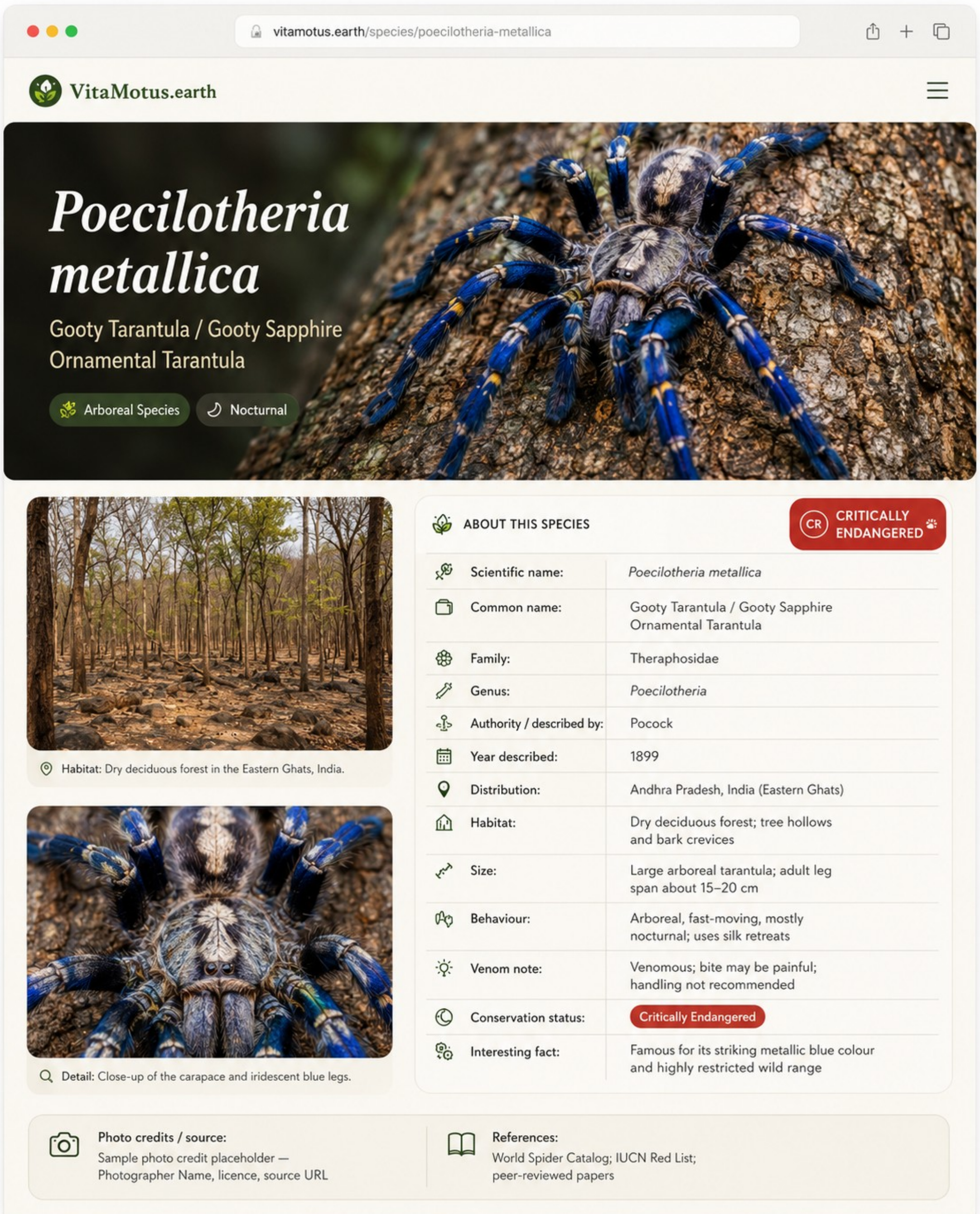
Species-card field	What it should contain
Scientific name	Full scientific binomial, for example Poecilotheria metallica.
Common name	Reliable common name or names, when available.
Family	Scientific family name, for example Theraphosidae.
Genus	Genus name, for example Poecilotheria.
Authority / described by	Person who described the species.
Year described	Year the species was described.
Distribution	Known country/region distribution.
Habitat	Natural habitat, such as dry deciduous forest, leaf litter, tree hollows, grassland, caves, etc.
Size	Body size or leg span where reliable.
Behaviour	Key behaviour such as arboreal, burrowing, web-building, nocturnal, fast-moving, social/solitary.
Venom note	Simple, responsible note on venom/medical relevance. No sensational language.
Conservation status	IUCN or other status if available; otherwise data deficient / not evaluated / unknown.
Interesting fact	One memorable fact that helps the visitor connect with the species.

Photo credits / source	Photographer, license, source URL, or placeholder until permission is obtained.
References	World Spider Catalog, papers, books, IUCN, regional sources, and other references used.

Species page look and feel: high-impact image panel, conservation banner, overview table, image credits, and references.

4A. Alternative Species Layouts

These mockups show alternate species page structures. The exact visual layout can change, but the standard fields must stay consistent.



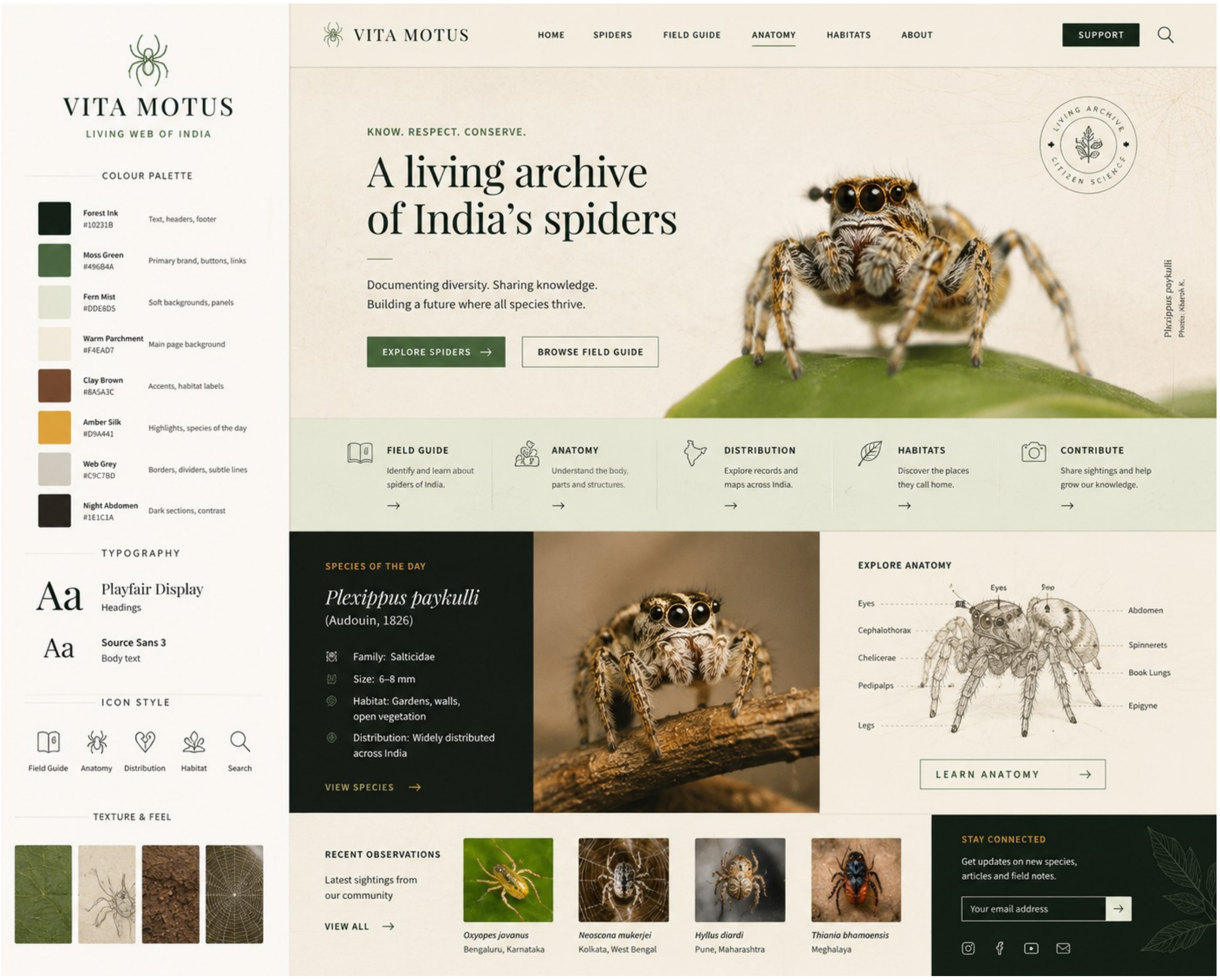
Alternative species layout: wide hero image, side gallery, structured overview card, conservation badge, photo credits, and references.

Additional species-card direction: split visual and data treatment for a high-impact profile page.

- Option 1: editorial profile with large left image and right-side data table.
- Option 2: wide hero image followed by structured cards and habitat/detail image panels.
- Option 3: compact species card for database browsing, linking to the full profile.

5. Taxonomy and Database Backbone

The first version should be a structured spider taxonomy archive, not a loose collection of isolated pages. The clean backbone is Family -> Genus -> Species.



Full site system reference: homepage, field guide areas, species cards, anatomy panel, and source-backed records.

Database layer	Developer meaning
Family table	Stores family_id, family_name, authority, year_described, accepted_genus_count, accepted_species_count, source, last_checked_date, notes.
Genus table	Stores genus_id, family_id, genus_name, authority, year_described, accepted_species_count, source, last_checked_date, notes.

Species table	Stores species_id, family_id, genus_id, scientific_name, genus_name, species_epithet, authority, year_described, distribution, profile_status, source, last_checked_date, notes.
Public views	Tree view for relationships, table view for search/filter, card view for normal visitors.
Admin import	Accept CSV/XLSX where one row equals one species. Family and genus pages should be generated from the species records.

Phase 1 should prioritise a reliable import and display system over decorative complexity. Once the skeleton exists, each species can gradually become a full educational profile.

6. Complete Spider Family Summary - All 138 Records

This table replaces the earlier species-count order list. It is alphabetical by family name and must stay complete in the guide. Columns: serial number, family, accepted genera count, accepted species count.

No.	Family	Accepted genera	Accepted species
1	**Actinopodidae**	3	128
2	**Agelenidae**	100	1482
3	**Amaurobiidae**	27	210
4	**Anamidae**	10	232
5	**Anapidae**	59	233
6	**Ancylometidae**	1	10
7	**Antrodiaetidae**	4	37
8	**Anyphaenidae**	59	663
9	**Araneidae**	197	3159
10	**Archaeidae**	5	93
11	**Archoleptonetidae**	2	8
12	**Argyronetidae**	12	74
13	**Arkyidae**	2	38
14	**Atracidae**	3	39
15	**Atypidae**	3	62
16	**Austrochilidae**	2	9
17	**Barychelidae**	39	294
18	**Bemmeridae**	4	52

No.	Family	Accepted genera	Accepted species
19	**Caponiidae**	21	157
20	**Cheiracanthiidae**	15	385
21	**Cicurinidae**	4	183
22	**Cithaeronidae**	2	10
23	**Clubionidae**	18	683
24	**Corinnidae**	77	897
25	**Ctenidae**	48	613
26	**Ctenizidae**	2	6
27	**Cyatholipidae**	23	58
28	**Cybaeidae**	24	308
29	**Cycloctenidae**	9	81
30	**Cyrtaucheniidae**	6	109
31	**Deinopidae**	3	71
32	**Desidae**	63	324
33	**Dictynidae**	45	343
34	**Diguetidae**	2	16
35	**Dipluridae**	8	155
36	**Dolomedidae**	7	129
37	**Drymusidae**	2	19
38	**Dysderidae**	24	671
39	**Entypesidae**	7	41
40	**Eresidae**	9	126
41	**Euagridae**	14	87

No.	Family	Accepted genera	Accepted species
42	**Euctenizidae**	8	79
43	**Filistatidae**	18	193
44	**Gallieniellidae**	5	41
45	**Gnaphosidae**	154	2500
46	**Gradungulidae**	8	20
47	**Hahniidae**	29	244
48	**Halonoproctidae**	6	150
49	**Hersiliidae**	16	189
50	**Hexathelidae**	7	45
51	**Hexurellidae**	1	8
52	**Homalonychidae**	1	2
53	**Huttoniidae**	1	1
54	**Hypochilidae**	2	33
55	**Idiopidae**	23	455
56	**Ischnothelidae**	6	28
57	**Lamponidae**	23	192
58	**Lathyidae**	10	58
59	**Leptonetidae**	22	400
60	**Linyphiidae**	641	4967
61	**Liocranidae**	35	358
62	**Liphistiidae**	8	199
63	**Lycosidae**	142	2542
64	**Macrobunidae**	27	95

No.	Family	Accepted genera	Accepted species
65	**Macrothelidae**	7	55
66	**Malkaridae**	13	57
67	**Mecicobothriidae**	1	2
68	**Mecysmaucheniidae**	7	25
69	**Megadictynidae**	2	2
70	**Megahexuridae**	1	1
71	**Melloinidae**	1	5
72	**Microhexuridae**	1	2
73	**Microstigmatidae**	12	44
74	**Migidae**	11	108
75	**Mimetidae**	8	167
76	**Miturgidae**	33	191
77	**Myrmecicultoridae**	1	2
78	**Mysmenidae**	17	188
79	**Nemesiidae**	10	199
80	**Nesticidae**	16	291
81	**Nicodamidae**	7	27
82	**Ochyroceratidae**	9	187
83	**Oecobiidae**	7	134
84	**Oonopidae**	115	1984
85	**Orsolobidae**	30	190
86	**Oxyopidae**	10	448

No.	Family	Accepted genera	Accepted species
87	**Pacullidae**	4	38
88	**Palpimanidae**	20	185
89	**Paratropididae**	6	48
90	**Penestomidae**	1	9
91	**Periegopidae**	1	3
92	**Philodromidae**	31	531
93	**Pholcidae**	98	2066
94	**Phrurolithidae**	27	427
95	**Physoglenidae**	13	72
96	**Phyxelididae**	14	68
97	**Pimoidae**	2	87
98	**Pisauridae**	45	235
99	**Plectreuridae**	2	32
100	**Porrhothelidae**	1	6
101	**Prodidomidae**	24	197
102	**Psechridae**	2	66
103	**Psilodercidae**	10	231
104	**Pycnothelidae**	15	146
105	**Rhytidicolidae**	2	16
106	**Salticidae**	695	6960
107	**Scytodidae**	4	263
108	**Segestriidae**	5	181
109	**Selenopidae**	9	282

No.	Family	Accepted genera	Accepted species
110	**Senoculidae**	1	31
111	**Sicariidae**	3	177
112	**Sparassidae**	99	1569
113	**Stasimopidae**	1	56
114	**Stenochilidae**	2	13
115	**Stiphidiidae**	20	125
116	**Symphytognathidae**	10	112
117	**Synaphridae**	3	13
118	**Synotaxidae**	5	40
119	**Telemidae**	16	113
120	**Tetrablemmidae**	27	154
121	**Tetragnathidae**	45	997
122	**Theraphosidae**	189	1194
123	**Theridiidae**	138	2623
124	**Theridiosomatidae**	24	182
125	**Thomisidae**	172	2198
126	**Titanoecidae**	5	67
127	**Toxopidae**	14	82
128	**Trachelidae**	30	322
129	**Trachycosmidae**	20	148
130	**Trechaleidae**	18	136
131	**Trochanteriidae**	6	51

No.	Family	Accepted genera	Accepted species
132	**Trogloraptoridae**	1	1
133	**Udubidae**	6	58
134	**Uloboridae**	20	284
135	**Viridasiidae**	3	14
136	**Xenoctenidae**	4	33
137	**Zodariidae**	90	1326
138	**Zoropsidae**	28	186

Footnote: Family names and species names use Latin scientific names. Where reliable, common English names can also be added for easier public understanding.

7. Growth Phases

Build a useful skeleton first, then enrich it steadily without losing quality control.

Phase	What to build
Phase 1 - Taxonomy skeleton	Build Family -> Genus -> Species pages. No long species write-ups required yet.
Phase 2 - Basic species profiles	Add common names where reliable, authority, year, distribution, habitat, size, behaviour, venom note, conservation status, references.
Phase 3 - Rich profiles	Add photos, range maps, identification notes, similar species, web type/hunting style, reproduction, and field notes.
Phase 4 - Hobbyist explanation layer	Add simple explanations: how to recognise it, where it is found, whether it is dangerous, and what makes it interesting.
Phase 5 - Interactive learning	Develop interactive spider anatomy, glossary, taxonomy visualisations, and beginner learning paths.

8. Developer Build Notes and Physical File Index

Use this section as the handoff checklist for the developer and as the index for the new physical project file.

Build note	Requirement
Backend source of truth	Relational database or structured CMS content model. Species row is the core record.
Import requirement	Developer should support CSV/XLSX import for family, genus, species, counts, status, source, and notes.
Dynamic Species of the Day	Cron/scheduled logic or server function chooses one species every 24 hours from eligible species records.
Profile status	Every species should have profile_status: pending, basic, draft, reviewed, or complete.
Citation fields	Every taxonomic record should support source and last_checked_date.
Image permissions	Public pages must use owned, licensed, permission-based, or properly credited images only.

- Physical file tab 1: Mission statement and website scope.
- Physical file tab 2: Website look and feel images.
- Physical file tab 3: Homepage and public page structure.
- Physical file tab 4: Species page standard template.
- Physical file tab 5: Anatomy and terminology ideas.
- Physical file tab 6: Taxonomy and database structure.
- Physical file tab 7: Alphabetical family summary - all 138 records.
- Physical file tab 8: Developer notes, future phases, and pending questions.